

Topic 6E: Alcohols

37. know that alcohols can be classified as primary, secondary or tertiary

38. understand the reactions of alcohols with:

- i oxygen in air (combustion)
- ii halogenating agents:
 - PCl_5 to produce chloroalkanes
 - 50% concentrated sulfuric acid and potassium bromide to produce bromoalkanes
 - red phosphorus and iodine to produce iodoalkanes
- iii potassium dichromate(VI) in dilute sulfuric acid to oxidise primary alcohols to aldehydes (including a test for the aldehyde using Benedict's/Fehling's solution) and carboxylic acids, and secondary alcohols to ketones

In equations, the oxidising agent can be represented as [O].

- iv concentrated phosphoric acid to form alkenes by elimination

Descriptions of the mechanisms of these reactions are not expected.

39. understand the following techniques used in the preparation and purification of a liquid organic compound:

- i heating under reflux
- ii extraction with a solvent in a separating funnel
- iii distillation
- iv drying with an anhydrous salt
- v boiling temperature determination

CORE PRACTICAL 5: The oxidation of ethanol**CORE PRACTICAL 6: Chlorination of 2-methylpropan-2-ol using concentrated hydrochloric acid**